Who Chooses, Who Delivers, Who Judges? A Functional Architecture for Distributed Public Spending

Working paper — v1.14 (July 2026; latest deposited version: v1.12, DOI 10.5281/zenodo.21252911). This version reports channel-separated, conditional simulation contrasts and retires the earlier compound value-per-budget multiplier as a calibrated effect (see §6 and Appendix E4); it rests on the architecture and the qualitative credit-versus-coverage mechanism. Revised through successive adversarial and author review cycles, documented in the repository's roadmap. It consolidates the companion computational program (Offermann 2026b): the deterrence-complementarity rule, the semi-open transition path, the budget-release rule, and machine verification with the human second instance; and the companion's two-layer separation of the macro categorization from the allocation profiles, under which the distributed arm is robust to a bad central categorization while the central arm is fragile to it.

Abstract

Public spending routinely asks one hierarchy to choose projects, execute them, and certify its own performance — the fusion where waste, capture, and unaccountable failure concentrate. This paper asks whether a bounded, legally authorized share of that machinery can separate those functions — the hand that chooses, the hand that spends, and the hand that certifies — and the information that drives them, while preserving legal authorization and public auditability.

We present **Core v0**, a fully specified, object-level architecture. Within legally authorized planning scopes, citizens direct a non-withdrawable share of an existing public budget to projects that must declare value claims, affected parties, milestones, and evidence contracts up front. Proposing, execution, evidence production, fiscalization, and custody are separated; funds move in tranches against reviewed milestone evidence, with retention and guarantees; executors neither choose nor pay their inspectors; and every consequential state transition is public.

Its animating idea is a **credit-versus-coverage** mechanism: when central ranking rewards claimable political credit, it can systematically underweight the diffuse, low-visibility benefits that a coverage-based distributed process still surfaces, albeit under voice bias. We held this idea to a public adversarial review of 43 attacks across five rounds — each integrated into the design or recorded as a bounded limitation. The evidentiary contribution is the architecture, the credit-versus-coverage mechanism, and separate conditional findings on selection, verified delivery, and administrative cost — not the large value-per-budget multiplier an earlier version reported. In the paper's sole confirmatory computation, a preregistered, selection-only gate gave the central competent, harm-aware appraisal at delivery parity; the distributed contrast stayed positive but did not clear the registered rebuild bar, so under the frozen rule we retire the multiplier (§6). Because that comparator switches off the modeled central harm-blindness and does not test delivery or agenda construction, it bounds calibration under the tested construction, not Core v0's architecture-wide value. A separate, **exploratory** four-scenario robustness extension then models the central selector as the evidence describes it — its *direction* on every axis grounded in the literature (not its magnitude fitted): near-blind to diffuse harm on the low-visibility long tail (Hayek 1945; Scott 1998; Olson 1965; Bandiera–Prat–Valletti 2009), projecting its own priors and over-rating visible benefits (Broockman–Skovron 2018; Flyvbjerg et al. 2003), and credit-tilted (Mayhew 1974; Arnold 1990). In that **source-motivated declared reference scenario** — not an empirical calibration — coverage-routed distributed selection recovers about 98% of the model's full-information greedy reference against the central's ~44%: a 54-point conditional model contrast, robust across the declared space and the realistic degradation of Core v0's universal budget *routing* (routing is architectural; effective independent information is not). The central narrowly overtakes coverage only by abandoning the declared premises (granting it the harm-sight the literature denies it) on a near-harmless world, while the *mirror* idealization of the distributed arm wins by a landslide. These are declared reference points from a stylized comparative-institutions model, not calibrated field effects; and the one sensitivity that materially shrinks the gap — never its sign in the declared range — is correlated/common-mode error in the coverage arm. A matched four-cell extension then crosses selection with delivery: in the declared PROBABLE world, the full selection-and-delivery architecture exceeds the status quo by about 58.6 points of the same greedy reference [95% conditional Monte-Carlo interval +58.0, +59.2]. A three-layer factorial leaves the full Core v0 diagonal positive in every named world, although individual layer attributions reverse sign in extreme worlds; planning's direction is anchored but its standalone magnitude is

deferred. Administrative cost does not run against the distributed arm: net-budget accounting leaves it roughly neutral under a conservative low-spread floor, and yields a declared, if modest, advantage once an asymmetric-cost scenario (the central's appraisal/prioritization bureaucracy ≫ a platform with near-zero-marginal fiscalization) is allowed. Elementary propositions give sufficient conditions for incentive-compatible disbursement and collusion-proof fiscalization under independence and corroboration assumptions.

This is an architecture-and-mechanism contribution, not an impact evaluation: no pilot has run; the simulations provide conditional model contrasts for selection, delivery, and administrative cost, but their units are uncalibrated and partial-equilibrium and do not identify target-domain treatment effects; and the empirical anchors are predominantly infrastructure and procurement — the milestone-verifiable, project-shaped cases where the evidence is clearest — while the architecture itself is general across the broader project-shaped, discretionary domain, not limited to infrastructure; that broader generality is a design claim, not yet empirically validated. What it offers is a concrete, criticizable, pilotable institutional design — and a disciplined account of exactly what its evidence does and does not yet support.

1. Introduction

Modern states concentrate three things that need not sit together: the authority to decide what public money is spent on, the capacity to execute that spending, and the power to certify that it achieved anything. When all three live inside one hierarchy, accountability comes down to it auditing itself (Bovens 2007). The consequences are predictable and documented across political economy — discretionary allocation, self-reported compliance, capture by concentrated interests, and citizen distrust — from regulatory capture (Stigler 1971; Laffont and Tirole 1991) to audit turned ritual (Power 1997) to the distributional coalitions that entrench themselves in stable political systems (Olson 1982).

These failures are not abstract: they are why the value never reaches the people it was raised for. For the public money that funds concrete projects — infrastructure, works, local programs — what ultimately matters is not how much is appropriated but how much effective value reaches people per unit spent: a project that is never built, or built badly, helps no one, however well-intentioned the allocation (Okun's (1975) leaky bucket carried water that never arrived). This is a bounded criterion for project-type public investment, not a theory of the whole budget or of why states tax: redistribution and equity are distinct purposes of public finance that this paper does not evaluate (§8).

The standard debate responds with quantity: shrink the state or grow it. Both positions treat it as a single object. This paper argues the tractable question is architectural, not one of size: *which layers of state activity still require central monopoly, and which can now be distributed — with better accountability than the status quo — now that digital coordination has collapsed the transaction costs that once justified hierarchy?* (Coase 1937; Hayek 1945; Williamson 1985). It is a question with a lineage — where decisions should sit when knowledge is dispersed — from the economic calculation debate (Mises 1920) to Hayek's dispersed knowledge and its institutional treatment in Sowell (1980): decisions should live where the knowledge is, disciplined by feedback rather than top-down command. Rights guarantees, legitimate force, macro-fiscal stability, and enforceable adjudication plausibly remain central; information processing, project-level resource allocation, service execution, evidence production, and auditing plausibly do not.

We make the argument as a concrete design, not a manifesto. Core v0 is a complete reference architecture for distributing one bounded layer — the allocation, execution, and verification of a legally mandated share of an existing public budget — developed to the level of named objects, state machines, and decision rules (a corpus of more than one hundred architecture documents, fifty-nine designed hypotheses, and forty-three adversarial reviews, all public). Citizens receive periodic, non-withdrawable allocation capacity in a civic wallet; projects pass through a parallel-closure lifecycle in which funding, independent fiscalization, evidence commitments, and beneficiary confirmation must all close before execution; the executor never selects or pays its own auditors — which removes the agency cost of self-monitored compliance (Jensen and Meckling 1976); money moves only on reviewed milestones, with retention and externally materialized guarantees; and every consequential state transition is recorded in a citizen-legible, expert-auditable trail.

The *executor* —whoever carries out the project, be it a public work, a personal-care service, or an educational program— can be a public agency, a municipality, a foundation, a cooperative, or a firm, as the implementing authority's rules define, under one condition: non-monopolistic, honest competition, with bankruptcies and consequences for whoever fails to deliver. And the share each citizen directs is set by a public, versioned formula the authority chooses; its simple option is *equal for all eligible citizens*, under which no one buys more influence with more money.

A concrete example. Suppose a municipality funds care programs for vulnerable elderly people. Today, between the peso allocated and the help that actually arrives lies a gap where value is lost. Under this architecture, each resident can direct part of their taxes to the program they prefer; the foundation, cooperative, or service that carries it out is paid in tranches, and only once a verifier it did not choose confirms the help arrived; and every step is public and auditable.

What makes a design exercise count as research is the rigor it is put through, under one transversal discipline: we evaluate every objection comparatively — feasible Core v0 against the institution that presently performs the same function, never against an omniscient, benevolent ideal. This blocks the nirvana fallacy in both directions (Demsetz 1969; Section 2). Our contributions are:

1. **Distributing the allocation layer.** The core design contribution: applying the functional-distribution principle to resource allocation — citizens direct their share directly, delegate it, or govern it with personal rules — instantiated as a complete reference architecture (Core v0).
2. **Formalization of the incentive mechanisms** (Section 5). We model milestone-gated disbursement as a principal-agent game and derive its incentive-compatibility condition; we model bribery of protocol-assigned fiscalizers and derive a collusion-proofness condition under k-corroboration; and we prove two design-relevant comparative statics: weak verification must be compensated with financial terms, and guarantees are counterproductive when detection quality falls below the cost of capital. Milestones, retention, guarantees, and reputational memory form the design's deterrence stack.
3. **Computational evidence** (Section 6). A dependency-free, seeded agent-based simulation of 10,000 citizens tests the architecture's behavioral assumptions under rational ignorance, limited discovery attention, and social-proof cascades. The results discipline the design: they support some claims, sharpen others, quantify the leverage concentrated in the scope-construction layer, measure a viable open construction of it, and carry the comparison end to end: from allocation to delivered social value per unit of budget — one relevant criterion for this bounded slice of public spending

(alongside distributional and rights constraints the model does not represent). In the model, selection and verified delivery compose multiplicatively (an accounting identity, not an independent finding); a matched selection-and-delivery extension gains ≈ +58.6 points of a greedy reference in the declared world [95% conditional-MC interval +58.0, +59.2]; a conditional three-layer attribution keeps the full Core v0 diagonal positive in every named world while planning's standalone magnitude is left unquantified; and net-budget accounting leaves administrative cost roughly neutral under a conservative low-spread floor, with a declared Core v0 advantage under an asymmetric-cost scenario. These are channel-separated, conditional model contrasts, not an institution-wide calibrated multiplier. Under the frozen pre-registered rule, a selection-only benchmark that granted the central competent, harm-aware appraisal at delivery parity left the distributed contrast positive but below the 0.05 rebuild bar; the earlier multiplier is therefore retired (§6; Appendix E4). Because that benchmark switches off the modeled central harm-blindness and does not test delivery or agenda construction, the decision governs calibration under the tested construction, not Core v0's architecture-wide value. The load-bearing contribution is the architecture, the qualitative credit-versus-coverage mechanism, and the separation of selection, delivery, and cost.

4. **Adversarial review as method** (Section 7). The architecture was attacked systematically: forty-three attack briefs grounded in the political-science, economics, and law literatures, each answered by a paired defense and resolved under an explicit integrate-or-bound rule, with resolutions propagated through the corpus (all except the manuscript-review round's D037–D040, whose corpus propagation is tracked) and the full review record public by construction. The loop is a structured self-critique, not external validation, and we say so; we propose it, and its terminating rule—pending independent external validation—as a preliminary methodological proposal for institutional design research.

A companion study measures three effects that extend this architecture: ablation of the deterrence stack (its terms are individually redundant but jointly indispensable), the feasibility of AI-based fiscalization, and the effect of separating macro planning from allocation (robustness to poor central planning — an earlier-apparatus contrast, subject to the same magnitude caveat as the compound above, not a calibrated effect).

Section 8 states limitations with the same care as results, because under our method they are results: each is a named, bounded residual risk.

2. The functional distribution principle

We analyze the state as a stack of functional layers rather than a single institution: (a) rights guarantees and legitimate force; (b) binding adjudication; (c) rule-making; (d) macro-fiscal management; (e) macro planning and agenda framing; (f) project prioritization and resource allocation to concrete projects; (g) execution and service delivery; (h) evidence production about delivery; and (i) evaluation and accountability. The distribution principle is:

> A layer is a candidate for distribution when three conditions hold: its failures under monopoly are information and incentive failures rather than coordination-of-force failures; distributed provision can be made *more* observable than monopoly provision; and the layer can be bounded so that its failure does not cascade into the non-distributable layers.

Layers (a), (b), and (d) fail the first or third condition and remain central in our design. Layers (f) through (i) pass all three, and Core v0 distributes them as a block, and must: distributing allocation without verification reproduces the delivery gap of participatory budgeting (PB), and distributing verification without allocation reproduces the audit society.

Layer (e), planning, is the deliberately unresolved case: Core v0 requires planning scopes to be public, versioned, and mandate-bearing, but does not distribute their construction — which is why the architecture's promise is stated with its qualifier built in: what it distributes is allocation *within mandate-bound, visible, versioned, and contestable planning scopes*, never allocation over an unframed agenda. Section 6 shows the qualifier is not a detail: it is the binding constraint of the whole design, and Section 8 elevates its removal to the research program's next object.

Two methodological rules govern everything that follows. The **comparative critique rule** (P001): every objection is evaluated against the current institutional baseline, not an ideal — a difficulty shared by both systems is a general problem of governance, not a refutation of the proposal. The **integrate-or-bound rule** (P007): once the core architecture is complete, a founded objection produces a new mechanism only if the failure mode would defeat a core claim and cannot be controlled through existing objects; otherwise it produces an explicit boundary, a visible residual risk, and a limitation statement. The first rule disciplines critics; the second disciplines the designers — a bias toward few, simple, general rules over proliferating specific machinery in the spirit of Epstein (1995).

3. Related work

Polycentric governance. Ostrom's demonstration that common-pool resources can be governed by nested, self-organized regimes without central monopoly (Ostrom 1990) is the closest intellectual ancestor: her design principles — bounded scope, monitoring by accountable monitors, graduated sanctions, cheap conflict resolution — reappear here as software-enforced objects. The difference is scope and instrument: we target budgeted state functions rather than natural-resource commons, and encode the regime in a platform whose rule changes are themselves versioned, auditable objects.

Participatory budgeting. Porto Alegre-style PB delegates a share of a municipal budget to citizen assemblies (Wampler 2007; Baiocchi and Ganuza 2017). Empirically, PB improves some fiscal outcomes but suffers from engagement decay, capture by organized minorities, and weak links between allocation and verified delivery (Peixoto and Fox 2016). Core v0 differs on exactly those margins: allocation is continuous and individual rather than assembly-based; delivery is bound to allocation through evidential contracts and conditional disbursement; and the architecture treats low participation as a design input (Section 6) rather than a failure to be exhorted away — collective action does not sustain itself at scale (Olson 1965). The civic wallet itself generalizes the voucher mechanism (Friedman 1962) — citizen-directed public money — from choosing among service providers to allocating among verifiable projects, adding the verification lifecycle that vouchers never carried.

Fiscal federalism and epistemic democracy. The formal ancestors of the functional distribution principle are the decentralization literature — Oates's (1972) theorem on when decentralized provision dominates, Tiebout (1956) on preference revelation through jurisdiction choice, and Besley and Coate (2003) on centralized versus decentralized provision under political economy — with one deliberate difference: our layers are functional rather than territorial, so what is distributed is a stage of the spending process (allocation, execution, verification) rather than a level of government. On the epistemic side, the aggregation results of Section 6 belong to the lineage of Condorcet's (1785) jury theorem and its modern failure conditions (Austen-Smith and Banks 1996) — a debt we make explicit, because the theorem's failure regime (correlated, strategic, or biased signals) is exactly what the seventh experiment stress-tests — and the design conversation with Landemore's (2020) open democracy and Fung and Wright's (2003) empowered participatory governance is direct: those programs distribute deliberation and empowerment; this one distributes allocation, execution, and verification with a mechanism-design and audit-trail core they do not attempt.

Liquid democracy. Transitive or scoped delegation promises flexibility between direct and representative participation, at the cost of concentration (Blum and Zuber 2016; Kahng, Mackenzie and Procaccia 2018). Core v0 adopts scoped, revocable, non-compensated delegation with mandatory concentration visibility, and — following Michels' (1911) warning rather than dismissing it — treats delegate concentration as a monitored risk with stress thresholds, not a solved problem.

Digital and blockchain governance. The DAO literature demonstrated both the feasibility of rule-encoded collective resource allocation and its characteristic failure: plutocratic token voting and governance capture (De Filippi and Wright 2018). Core v0 is deliberately not a blockchain design — identity is verified rather than pseudonymous, and the sovereign state remains the source of funds and law — but it imports the lesson that meta-governance is the highest-leverage attack surface (Section 8).

Mechanism design and audit. Our formal models are elementary applications of moral hazard under imperfect observation (Holmström 1979) and collusion in supervision hierarchies (Laffont and Tirole 1991), with Goodhart's law (Goodhart 1975; Campbell 1976) as the standing warning against metric gaming. The closest existing mechanism design for citizen allocation of public funds is quadratic (or 'plural') funding (Buterin, Hitzig and Weyl 2019), which prices concentration through matching-fund curvature; Core v0's funding-target closure rule pursues the same anti-concentration goal by truncation rather than pricing, and our simulation results (Section 6, Finding 1) delimit what truncation can and cannot achieve. On the audit side, Olken's (2007) field experiment on Indonesian road projects is the canonical empirical anchor for the detection probabilities our Propositions 1–2 take as parameters — and its finding that top-down audit outperformed grassroots monitoring for procurement fraud is a caution this architecture absorbs by making professional fiscalization, not crowd observation, the release-gating layer. The Brazilian audit-lottery literature (Ferraz and Finan 2008) supplies the complementary mechanism evidence — disclosure of audit findings changes political outcomes, and audit exposure reduces subsequent corruption — and its underlying CGU data enter the seventh experiment's audit-parameterized baseline directly. The contribution here is not technical depth but specificity: the models' parameters map one-to-one onto named architectural objects, so every proposition is an implementable dial.

What is new. We have not run the systematic prior-art review that would establish field-level priority (our literature map is preliminary), so we claim an **integrative, object-level synthesis** rather than novelty against all adjacent work. With that qualification, we are not aware of prior work combining:

- **(i)** a functional decomposition of state activity into distributable and non-distributable layers;
- **(ii)** a complete object-level architecture for the allocation layer;
- **(iii)** formal incentive analysis of that architecture's specific mechanisms;
- **(iv)** behavioral simulation of its citizen-facing assumptions —including what we believe is an early symmetric-knowledge comparison, in simulation, of open construction of allocation priorities from

aggregated citizen signals against bandwidth-constrained central construction (see Section 6);

- **(v)** a documented adversarial-review method with an explicit stopping rule.

And two further contributions concern measurement and method:

- **(vi)** an end-to-end institutional comparison, within a bounded public-investment slice, on delivered social value per unit of budget, decomposing selection from delivery on matched portfolios: in the model, selection and verified delivery compose multiplicatively (an accounting identity, not an independent finding); a matched selection-and-delivery extension gains ≈ +58.6 points of a greedy reference in the declared world [95% conditional-MC interval +58.0, +59.2]; a conditional three-layer attribution keeps the full Core v0 diagonal positive in every named world while planning's standalone magnitude remains unquantified; and net-budget accounting leaves administrative cost roughly neutral under a conservative low-spread floor, with a declared Core v0 advantage under an asymmetric-cost scenario. Separately, a later pre-registered symmetric gate — a **selection-only** test with delivery held at parity — finds the *selection* advantage positive but small (Section 6); it does not test the delivery interaction. This work also introduces the visibility gap (officially reported minus real delivery) as a measurable accountability deficit of the status quo;

- **(vii)** that comparison against an audit-parameterized baseline built from the published findings of supreme audit institutions across nine jurisdictions and the European Union (a documented-practice-informed scenario whose primary-source verification is still being completed), under a pre-registered headline-withdrawal condition.

Participatory-budgeting evaluations measure participation and allocation; audit studies measure leakage after the fact; we know of none that measures, within one framework, how much delivered value an allocation institution produces from the same resources.

4. The Core v0 architecture

We summarize the reference architecture at the level needed for the analysis; the full object model, state machines, and citizen-interface layers are specified in the public corpus.

Funding. An implementing authority migrates a mandated share of an existing budget into individual civic wallets: periodic, non-withdrawable, public-purpose allocation capacity, equal per citizen by default. Every active planning scope carries an *Allocation Mandate* record naming the statute or instrument that authorized the migration, its legal rank, the organ to which allocations are imputed, and the allocation formula. The platform records that external authorization; it does not manufacture it. Binding-mode operation is gated on an enabling norm of sufficient rank being recorded; otherwise, the disclosed lawful default is consultative or tutored operation. The allocation act is designed to replicate two guarantees of the vote: secrecy of the preference and coercion-resistance (receipt-freeness). To the extent an enabling norm recognizes it, it can be shielded with protections equivalent to the vote's; until then, these are technical platform guarantees, not a legal status. Individual allocations are pseudonymous at the public layer and reconcile cryptographically against public per-scope totals — every peso traceable as money, no citizen traceable as an allocator, and no receipt or exportable proof of any individual allocation exists, even voluntarily, so that a patron who demands proof can never get one (the secret ballot's own defense, applied to the wallet). A *Fiscal Commitment Profile* per scope makes the migrated percentage, indexation, and delivery latency public and versioned, so fiscal strangulation by the incumbent treasury is measurable and attributable rather than silent. Essential services with continuity obligations are protected by non-assignable floors outside citizen-by-citizen popularity.

Projects and roles. Financeable projects declare a value thesis with verifiable claims, affected parties, risks and anti-values, a phase and milestone plan, and an *evidential contract*: what must be proven, by what class of qualified producer, with what method, for which formal effect. Six roles are structurally separated — proposer, modeler/designer, executor, fiscalizer, evidence producer, custodian — with related-party relationships declared on a severity-classified graph. The load-bearing rule is that the executor never chooses or pays its own fiscalizers or evidence producers: control work is financed from a separated control budget and assigned by protocol.

Parallel closure and conditional disbursement. A published project gathers funding commitments, fiscalizer assignments, evidence commitments, and beneficiary confirmations concurrently; execution becomes possible only when all conditions required by its proportional *threshold policy* close. Committed funds are custodied, not transferred: release happens per milestone, against reviewed fulfillment evidence, with retention, blocker checks, and guarantees materialized by external custodians before any release. A *Duty-of-Care Anchor* names, before disbursement, the solvent legal person civilly answerable to third parties for damages arising from execution, in particular damage to physical integrity.

Attention infrastructure. Citizens act through a layered interface: discovery with user-controlled, reason-visible ordering; compact project cards; and progressively deeper audit surfaces down to the full trail. Non-attending citizens are served by configurable automatic allocation profiles — or a sensible default profile when none is set — and by scoped, revocable delegation with concentration visibility. The architecture does not assume attentive citizens; it assumes mostly inattentive ones and routes their weight through inspectable intermediation (Lupia and McCubbins 1998). This is a design answer to the citizen-competence objection in its sharpest contemporary form (Brennan 2016): rather than restricting anyone's right to participate, the architecture makes the intermediation that inattention produces visible, revocable, and auditable.

An apparent objection —that participating via app, wallet, and AI tutor excludes the non-digital population— dissolves under the comparative discipline: the non-digital citizen already delegates today, handing their decision, through the vote, to a distant representative who allocates the budget for them. Core v0 does not add a barrier: it removes a level of indirection. Whoever never participates falls to the system default —equal per citizen, mandate-bound—, not to the attentive minority's preference; and whoever participates even minimally, including through non-digital channels or assisted delegation, brings the decision closer to their direct interests through microdelegation and rules such as "near me" that fund what they can touch. What appears to exclude, includes more —with the construction of the planning scope as the only remaining indirection (Section 8).

Transition. Deployment proceeds through operating modes — closed, tutored, semi-open, open — in which a public authority may retain eligibility review (project admissibility), but every material tutored decision, and every tutored silence past its deadline, becomes a public governance-resolution object. Incumbent-resistance indicators (scope share opened, rejection and timeout rates, operator privilege) make symbolic adoption distinguishable from real transfer.

5. Formal analysis

We state the three models and their results; proofs are one-step algebra and appear in the companion note (formal-models). All agents are risk-neutral; budgets are normalized to 1. The deterrence structure throughout is Becker's (1968): a violation is deterred when the detection probability times the stake at risk exceeds its gain — our contribution is mapping each term of that inequality onto a named, configurable architectural object. To avoid notation collisions, *Proposition N* denotes the formal results of this section; *P001/P007*, the methodological rules (§2); and *prediction N*, the behavioral predictions of §5.3.

5.1 Milestone-gated disbursement

An executor chooses to deliver a milestone at private cost $c \in (0, 1)$ or to divert. The mechanism releases an advance a, holds the remainder for reviewed acceptance, recovers a fraction r of the advance on confirmed non-delivery, seizes a posted guarantee γ (carrying cost κ per unit), and imposes a reputational continuation loss R. Review confirms diversion with probability p — the joint quality of evidence standards, fiscalizer independence, and corroboration.

Proposition 1 (incentive compatibility). Delivery is optimal iff

> $c \le p \cdot [\ (1 - a(1 - r)) + \gamma + R\ ].$

Delivery must be cheaper than the detection probability times the total stake at risk. Every disbursement-gaming test in the architecture is a term in this inequality.

Proposition 2 (weak verification must be priced). The minimal guarantee sustaining delivery for all $c \le \bar{c}$ is $\gamma^*(p) = \max\{0, \bar{c}/p - (1 - a(1 - r)) - R\}$, decreasing and convex in p. Where verification is weak — thin fiscalization markets, poor evidence quality — the mechanism must compensate with smaller advances, more recoverability, larger guarantees, or higher-reputation executors. A single global guarantee percentage cannot be optimal across heterogeneous verification environments.

The deterrence terms of this condition are complements, not substitutes. A pre-registered ablation on the companion experiments (Offermann 2026b) measured the consequence: at the designed operating point the inequality holds with slack, so removing any single term costs almost nothing — and a deployment negotiated one defensible concession at a time can cross the threshold invisibly, ending below the status quo it replaced (a material shortfall in verified value, with selection quality intact) while looking fully instrumented. The corpus therefore requires every scope to publish its deterrence-inequality margin in its operating configuration, recomputed on every term change, with term reductions classified as material rule changes (docs/111). The same slack, kept intact, buys an unexpected dividend: it absorbs verification- instrument error — in the companion panel of five real model families, even a machine verifier passing ~20% of judgment-requiring fraud produced leakage indistinguishable from pure-human verification, because the cascade removes the attempts upstream — provided the adversary is not colluding across the verification layers (Offermann 2026b, docs/113).

Proposition 3 (participation-deterrence trade-off). Raising γ relaxes incentive compatibility at rate p but lowers honest executors' payoff at rate κ; under a designer objective weighing one unit of incentive slack equally against one unit of honest-executor payoff, a guarantee increase is net-beneficial only if $p > \kappa$ (other weightings shift the threshold, not the trade-off's structure). Where detection quality is below the local cost of capital, piling on guarantees excludes honest low-margin executors without deterring fraud — the formal content of the architecture's proportionality discipline.

5.2 Collusion-proof fiscalization

A non-delivered milestone is worth G to the executor if fraudulently approved. Release requires approval by k protocol-assigned fiscalizers, each carrying forfeitable stake W (future protocol fees plus role reputation) and facing post-approval discovery probability q. Because assignment is protocolized and repeat pairings are visible, executor and fiscalizer are strangers: a bribe offer is itself reported with probability δ, costing the executor penalty P_e.

Proposition 4 (collusion-proofness). Approval fraud is unsustainable if

$$k \cdot q \cdot W \geq G - (\delta / (1 - \delta)) \cdot P\_e,$$

and in particular whenever $kqW \geq G$. Three corollaries carry design weight:

- ***Corroboration substitutes for reputation capital.*** The required stake per fiscalizer falls linearly in k, so redundant review is exactly what makes shallow-reputation fiscalizer pools workable, at linear control cost —which is what proportional threshold policies are for.
- ***Repeat relationships are the attack surface.*** The friction term exists only while relational contracting is prevented, which is why repeat-pairing visibility is load-bearing (we hold the report probability δ exogenous to k; endogenizing it —more approached reviewers, more chances of a report — only strengthens the condition).
- ***Thin markets attack both models at once.*** A monopolist fiscalizer that cannot credibly be excluded loses its forfeitable stake ($W \to 0$) while also degrading p in Proposition 1: the two conditions identify the same environments as fragile, for the same reason.

5.3 Attention-constrained allocation

Citizens allocate small individual budgets; the pivotal return to evaluating projects is negligible, so rational ignorance is the equilibrium (Downs 1957), and the design question is where the *inattentive* majority's weight flows: to salience amplified by social proof (Bikhchandani, Hirshleifer and Welch 1992; Noelle-Neumann 1974; Salganik, Dodds and Watts 2006), or to the architecture's own default layer, whose routing follows the distributed project prioritization — the aggregated allocation profiles — not a central plan. The model yields three testable predictions — caps tame cascades (prediction 1), defaults anchor quality (prediction 2), decay degrades gracefully only with defaults (prediction 3) — evaluated next.

6. Computational evidence

We test the three predictions of §5.3 —and, in successive experiments, the assumptions of Propositions 1–4— in an agent-based simulation. Each experiment (E1–E10) corresponds to a finding:

Exp	What it tests	
E1	do funding caps raise quality?	Finding 1
E2	what carries allocation quality?	Finding 2
E3	what buffers participation decay?	Finding 3
E4	distributed aggregation vs. central construction (refined by a symmetric-frictions frontier + capture, E4-v4/v5; and a v1.14 harm-myopia four-scenario robustness map, §6)	Finding 4
E5	delivered value: selection × delivery, at matched budget	Finding 5
E6	reputational competition and execution standard	Finding 6
E7	comparison against an audit-parameterized baseline	Finding 7
E8	robustness under endogenous behavioral participation	Finding 8
E9	the full stack: planning × selection × delivery (Shapley attribution)	Finding 9
E10	the administrative-cost layer (net-budget, symmetric)	Finding 10

We simulate 10,000 citizens over 24 monthly cycles allocating across a standing pool of 40 projects with quality θ, salience s (measured corr(θ, s) ≈ 0.24), prioritization need-weights $w = \lambda\theta + (1 - \lambda)u$ (where u is an idiosyncratic need component independent of quality) with mixing weight $\lambda \in \{0.4, 0.8\}$ — measured corr(θ, w) ≈ 0.55 and ≈ 0.97 respectively — and 3× scarcity (only a minority of projects can complete). Evaluators (2–10%) fund the best quality they sample; salience followers see a six-slot discovery surface ranked by salience amplified by funding progress; default followers' budget fills projects in planning- priority order. The funding-target closure rule is toggleable. Twenty seeded runs per condition; the code is dependency-free and deterministic (`scripts/simulation/allocation-sim.mjs` ; full tables in simulation-results).

Quantitative status: channel-separated results, no calibrated multiplier. The earlier single value-per-budget ratio conflated selection quality, delivery leakage, and administrative cost and is **retired as a calibrated effect**. The rebuilt program reports the channels separately: selection (E4), delivery at matched budget (E5), a conditional three-layer attribution (E9, leaving planning's standalone magnitude unquantified), and administrative cost (E10). All are reference-point differences with parity at zero; no institution-wide multiplier is retained.

The governing pre-registered decision. The paper's **sole confirmatory** computation was a selection-only, symmetry-constrained stress test — not a test of the full architecture. It matched the candidate pool, costs, budget, report-noise model, own-estimate eligibility, expected appraisal-report budgets, and delivery at parity. The distributed arm retained endogenous coverage and its adverse voice bias β; central appraisal was spread evenly and its ranking retained bounded credit pressure $\lambda \in \{0.1, 0.2, 0.3\}$ (the separate $\lambda = 0$ control is even smaller, median ≈ 0.016 — the contrast tracks the central's credit pressure, consistent with credit-versus-coverage). The implemented central was a **competent, harm-aware value reader** whose sampled-value signal was unbiased but noisy: the harm-myopia parameter is defined in the world parameters but **unused by the central estimator**. The gate therefore switches off the modeled central harm-blindness and holds agenda construction outside its fixed-pool estimand.

The distributed contrast was **positive in all 18 primary cells**. Three of the four conjunctive criteria passed, including the bootstrap lower-bound criterion; only C2 failed, because the pooled median $\Delta = 0.025$ did not clear the registered **0.05 rebuild bar**. Under the frozen rule the formal verdict was **NO-GO**, with the registered consequence that the quantitative multiplier is retired and the simulation treated as an illustrative conditional frontier. **The threshold miss triggered the registered decision; the construction determines its scope.** The 0.025 is small for the tested selection estimand, but it is neither an estimate nor a lower bound of Core v0's architecture-wide or target-domain effect, and it does not establish smallness outside this benchmark. The 0.05 bar is a research-program decision rule, **not** a calibrated policy-materiality threshold. The hypothesized harm-detection asymmetry is literature-motivated; this study does not calibrate its prevalence or magnitude. The subsequent four-scenario map uses a different data-generating process, is **exploratory**, and does not revise this registered result. The test is stylized: its value and credit variables are abstract scores, not measured visibility, traceability, permanence, or public value. Its frozen pre-registration, decision rule, results, and diagnostics are in Appendix E4, the claim-and-estimand contract, and `scripts/simulation/e5-sp-symmetry-gate.mjs`. The load-bearing contributions are the architecture and the qualitative credit-versus-coverage mechanism.

The declared reference scenario (E4 v1.14) — an exploratory robustness map. When the central selector is modelled as the evidence describes it — its *direction* on every axis grounded in the literature (not its magnitude fitted): near-blind to *diffuse harm on the low-visibility long tail* (Hayek 1945; Scott 1998; Olson 1965; Bandiera, Prat and Valletti 2009, whose **83% passive waste** in standardized-goods procurement shows most public loss is not chosen), projecting its own priors and overrating visible, appraisable benefits (Broockman and Skovron 2018; Flyvbjerg et al. 2003), and tilted toward claimable credit (Mayhew 1974; Arnold 1990), in a heavy-tailed low-visibility project space (Skuhrovec et al. 2013) — **coverage-routed distributed selection recovers about 98% of the model's full-information greedy reference against the central's ~44%: a 54-point conditional model contrast**, not an empirical calibration or a field effect (the reference is a greedy normalization, not a feasible institution or a welfare optimum). It is not a knife-edge: it holds across the declared parameter space; it survives the realistic signal degradation of Core v0's universal budget *routing* (a ~~5% active-report / ~35% microdelegation / ~60% profile-rule composition — universal *routing* is architectural, effective independent *information* is not); and it holds even when the central is granted the full harm-aware competent bundle the literature would deny it — harm-sight, unbiasedness, precision, and no credit (~~+14%). The central narrowly overtakes coverage only by **further abandoning the declared premises** — also moving to a near-harmless world where the diffuse harm the literature documents barely exists; the *mirror* idealization of the distributed arm (its own recipe reflected: perfect signal, harm-rich world, central kept at its declared myopia) wins by a landslide (~+118%). The single sensitivity that materially shrinks the gap — without reversing its sign in the declared range — is correlated/common-mode error in the coverage arm (a shared platform/recommender or concentrated delegation), which brings ~+54% to ~+26% at a strong correlation. This is the mechanism's one architectural condition, not merely a swept parameter: **coverage without source diversity can reproduce the epistemic bottleneck it is meant to replace.** Core v0 therefore treats delegate, profile-provider, and recommender concentration as observable quantities with diversification thresholds (§8), rather than assuming independence by fiat. The pre-registered symmetry test remains the paper's separate **confirmatory** computation (a NO-GO near-parity on a symmetric benchmark, under a different data-generating process); this map is a subsequent exploratory scenario analysis, not a reclassification of it. The full four-scenario map, the literature anchoring, the mirror idealized corners, and the common-mode frontier are in **Appendix E4**. The model locates a frontier; it does not estimate a field effect.

Finding 1: funding caps are an anti-concentration device, not a quality device. With closure ON, concentration falls (funding Gini 0.732 vs 0.759), the most salient 5% of projects absorb less (16.8% vs 19.6% of all funding), and 15% more projects complete (25.3% vs 21.9%). But quality selection is unchanged (*sel(θ)*, the Pearson correlation across projects between latent quality θ and the binary indicator that a project reaches its funding target, ≈ 0.39 vs 0.41): the truncated surplus spills to the next most *visible* project, not the next *best* one. The architecture's claim for the closure rule should be — and in the corpus now is — bounded accordingly.

Finding 2: the default anchor, not citizen attention, carries allocation quality. A default-anchored mix, with a near-perfectly informed planner (r ≈ 0.97), reaches sel(θ) ≈ 0.71 — well above the salience-driven configurations (≈ 0.35–0.43). Raising citizen attention from 2% to 10%, by contrast, barely moves the needle —at most ≈ 0.08 in salience-driven regimes, and essentially nothing in default-anchored ones—, while degrading the vector's informational quality from near-perfect to moderate (r ≈ 0.97 → 0.55) costs ≈ 0.29 of selection. The anchor rules, not attention.

Two qualifications keep the finding honest:

- **By construction.** The default rule is a deterministic θ-correlated allocator already holding most of the budget; what is measured is the *conditioning* —how much the vector's informational quality determines the anchor's value, and how little attention substitutes for it—, not the wisdom of the crowd.
- **Robustness.** A sensitivity panel (varying evaluator sample size and social-proof strength) shows the regime ordering is robust, except under very strong social proof, where the regimes converge within noise because strong amplification also propagates the evaluators' quality signal. Magnitudes are parameter-dependent and uncalibrated; what survives all variations is the ordering and the dominance of the prioritization's informational quality.

This finding quantifies the leverage concentrated in whatever constructs the project prioritization the passive share follows. That prioritization has two layers —which a companion study (Offermann 2026b) separated for the first time—: the macro categorization (this corpus's Planning Scope, which frames eligibility and carries no budget weights) and the aggregated allocation profiles that route budget within it. The distributed arm is robust to the quality of that categorization and a central arm is fragile to it, so the advantage over a central status quo is not fixed: it grows as central planning worsens — a model-internal direction the companion apparatus illustrates (a conditional contrast that widens substantially as the central categorization degrades, not a calibrated multiplier; see the quantitative-status note in this section).

Two architectural facts scope the statement and forestall a tempting over-reading. First, the default layer is pluggable, not mandatorily central: the civic autopilot gives each citizen manual allocation, delegation, published allocation profiles, a personal automatic rule, or the system default; an onboarding citizen must explicitly select or acknowledge a base profile, and only the share who never engage necessarily follows the system default —which itself operates under a recorded allocation mandate. Second, centralized construction of scope weights is a property of the closed and tutored transition modes, not of the architecture: operating modes are country-configured states, and the designed trajectory is toward open construction (Finding 4 measures its in-model viability).

The numbers therefore establish a conditional. The binding constraint on allocation quality is the informational quality of the **project prioritization the passive share follows** —the aggregated allocation profiles, not a macro planning vector—, whoever or whatever supplies it. A captured or ignorant supplier is the failure mode; a well-informed or well-aggregated one, the asset. Randomizing that prioritization to escape capture does not help: it buys neutrality at the price of near-random quality for the passive share. And because the architecture's designed trajectory distributes its construction (open mode) and keeps it visible, versioned, and pluggable, the constraint is met by distribution, not by a central agenda. This is distinct from the narrower agenda-setting point of Section 8, which concerns only who frames eligibility.

E1–E3 do not authorize two readings: the prioritization's origin is unspecified (r is a property of the vector, not of a state office), and the modeled crowd carries social proof but no knowledge —so these experiments compare attention against weight quality, not central against distributed knowledge. Finding 4 makes that comparison properly.

Finding 3: what buffers participation decay is the anchor's level, not where the leavers' weight flows — our own prediction failed here. We predicted that decaying active evaluation (10% to 2% over 24 cycles) would degrade allocation gracefully only if the released share flowed to defaults rather than salience. Exploiting common seeds across conditions, the paired analysis rejects this: the destination's effect on overall selection is null at both anchor strengths (mean paired differences 0.001 [−0.031, 0.033] and 0.021 [−0.028, 0.071]), and the only interval excluding zero is small and opposite-signed (at a strong anchor, the salience destination preserves late-cycle alignment slightly better — plausibly because amplified social proof also propagates evaluator seeding). What governs robustness to decay is the structural share of the default layer itself, Finding 2's variable. Engagement decay — the documented fate of civic-tech participation (Hirschman 1970; Peixoto and Fox 2016) — is a buffered risk here, but the buffer is the institutional layer's size, and within-cycle quality alignment still erodes in later cycles under all conditions, so decay is bought, not free.

Finding 4: aggregated dispersed signals outperform fixed-bandwidth central construction of the weight vector — but a fair, symmetric re-examination resolves the advantage into a conditional frontier and adds a capture-resistance that guards it. A fourth, pre-registered experiment (design and predictions committed before any run; `research/e4-institutional-knowledge-design.md`) models knowledge symmetrically instead of endowing it: a planner with fixed bandwidth (thirty deep inspections; its correlation with latent quality is now measured output, collapsing 0.81 → 0.37 → 0.17 as the project pool grows 40 → 200 → 1000) against thirty percent of citizens holding four individually poor local signals each (noise 0.35). Five regimes share the identical world and signals and differ only in the aggregation institution. The pre-registered scale-crossover prediction failed in the informative direction: open construction of the weight vector — a plain average of citizen signals per project — beats pure central construction at *every* scale, including the smallest, where the planner inspects three-quarters of the world (sel(θ) 0.76 vs 0.62 at N = 40; 0.73 vs 0.04 at N = 1000). Twelve thousand noisy signals average into a near-perfect vector; thirty good inspections cannot compete, and fixed central bandwidth decays toward randomness as the world grows — a Condorcet (1785) aggregation logic, and we treat it as such: the jury theorem's known failure conditions (Austen-Smith and Banks 1996) define exactly the boundary the seventh experiment tests. Three qualifications carry the finding's honest weight. First, the same dispersed knowledge *without* an aggregation institution is wasted: the uncoordinated regime funds 0.6–15% of projects and selects poorly — the result vindicates aggregation mechanisms, not the absence of mechanism, and Core v0's default-plus-closure layer is exactly such a mechanism. Second, aggregation defeats noise, not bias: signals are unbiased by construction, and a common-mode bias uncorrelated with quality (misinformation, expressive allocation) would not average out — only endogenous social-proof contamination was tested, and proved largely benign because visible funding progress is itself quality-correlated in a well-anchored system. Third, elicitation is non-strategic by assumption; in deployment, signal reporting becomes a gaming and clientelism surface, and the mechanics of open scope construction remain a gated design problem. Within those bounds, the finding points in a clear direction: the binding variable is not who holds the pen but how much dispersed information the scope-construction institution ingests.

The simulation also disciplines rhetoric — in both directions. Nothing in E1–E3 supports describing Core v0 allocation as "the wisdom of crowds": the honest description is *inspectable intermediation with a citizen-correctable default*, which the results show is both realistic and better than the salience-driven alternative that unstructured platforms converge to. And nothing in E1–E3 licenses the opposite reading — that central planning knowledge beats distributed knowledge — because they never modeled distributed knowledge; when E4 does, aggregation wins wherever its named preconditions hold. Both discourses lose their slogan; the design keeps its numbers.

A fair, symmetric re-examination (E4-v4/v5). This first-pass E4 gave the central a fixed inspection bandwidth and left citizen signals unbiased — two idealizations that, an adversarial review showed, tilt the comparison in the distributed arm's favor. A rebuilt model (`scripts/simulation/e4-v4-symmetric-frontier.mjs`, `research/e4-v4-symmetric-frontier-results.txt`, `research/e4-v5-capture-design.md`) gives *both* institutions a symmetric friction in perceiving true value, including harm: the central attenuates perceived harm by a coefficient η (0 = blind to diffuse harm, 1 = a fully accountable planner), while the distributed reads true valuations but the diffusely harmed under-participate at a rate β (voice inequality). The result is not a multiplier but a frontier with a closed-form parity locus (`research/e4-analytical-backbone.md`): both institutions are biased estimators of the same Samuelson value $T = S^{+} - S^{-}$, ranking projects by $S^{+} - \theta \cdot S^{-}$ with θ_C = η and θ_D = 1 − β, so the distributed dominates exactly when its coefficient is closer to the true harm-weight of one — i.e. **β < 1 − η**. The simulation confirms the law (parity on the anti-diagonal η + β = 1) and quantifies the delivered-value degradation off it (from parity on the anti-diagonal to a substantial distributed lead across the plausible box). The advantage is thus a property of *including the harmed*, not of aggregation per se; it reaches parity along the anti-diagonal β = 1 − η and turns into a central win below it (β > 1 − η) — which absorbs the participation-bias objection into the model's own β axis rather than leaving it external. Neither extreme is assumed: η is *swept*, not fixed, and a low but non-zero η is a defended regime, not a premise. The diffuse-harm literature (Bastiat's (1850) unseen costs; Olson's (1965) asymmetric organization on contested issues; Wilson's (1973) client-politics quadrant; Scott's (1998) legibility) describes *when* diffuse costs go unrepresented — each read at its proper scope, not as a claim of global blindness

— while the opposing thesis that political competition disciplines the centre toward efficiency (Wittman 1989) holds weakest in exactly that client-politics quadrant. Empirically, most *measured* procurement loss is passive — incompetence, not theft (Bandiera, Prat and Valletti 2009) — consistent with a low but non-zero η.

Capture-resistance guards the advantage (E4-v5). Modelling organized capture symmetrically — the review's hardest objection, applied in fairness to the central planner too — the asymmetry widens rather than closes. A group captures a low-value project only when its private rent exceeds acquisition cost plus the expected sanction (Becker 1968); with the deterrence asymmetry carried entirely by detection probability and acquisition scaling (the penalty held equal, conservatively), the status quo turns net-harmful at rents near 10% of project cost while the distributed threshold — floored by the equal-per-citizen wallet, which money can persuade but cannot buy — sits substantially higher (closed form in the backbone note). Detection is a snowball $p = 1 - (1 - q)^m$, so its floor is an expected $m \cdot q \geq -\ln(1 - p_c) \approx 0.1$ reporters drawn from the transparent affected public — low, but this is a model-internal statement whose force depends entirely on the stipulated detection gap (central ~0.10 vs distributed ~1.0), not an empirical burden-of-proof: sensitivity analysis is decisive here — the distributed advantage narrows and can reverse if distributed detection is driven down toward ~0.3, so the claim is that organized recapture is harder under the distributed arm's transparency *given these parameters*, not that it is ruled out. This ties Finding 4 to the integrity layer of Finding 5: the same fiscalization that makes value arrive is what keeps the allocation advantage from being bought back by organized rents, so the two are a layer and its safeguard rather than independent multipliers. Every magnitude here is model-internal; the literature (Olson 1965; Wilson 1973; Scott 1998; Bastiat 1850; Becker 1968; Becker and Stigler 1974; Stokes 2005; Dyck, Morse, and Zingales 2010; Ostrom's 1990 self-monitoring commons) defends the direction, the mechanism, and the sign of the asymmetry — not the numbers.

Analytical backbone. Three closed forms carry the weight, each verified against the simulation (`research/e4-analytical-backbone.md`); the runs then only confirm them and quantify the degradation off the clean cases. **(i) The parity law.** Writing each institution as a biased estimator that ranks projects by $S^+ - \theta \cdot S^-$, the central keeps $\theta_C = \eta$ of perceived harm and the distributed reveals $\theta_D = 1 - \beta$ (the participation rate cancels from the ranking); since the true harm-weight is one, the distributed arm delivers more true value **iff $\beta < 1 - \eta$**, parity on the anti-diagonal. A bias–variance reading would predict that on the parity line, where the bias cancels, the lower-variance estimator wins — the distributed's revelation noise is zero (a funder knows her own value), the central's proxy noise is not. The implemented simulation does **not** bear this out at the accountable corner: at $(\eta = 1, \beta = 0)$ the measured outcome is a slight **central win** (the distributed falls just short of the central there) — so the honest reading is the noise-free parity law $\beta = 1 - \eta$, and the bias–variance tilt toward the distributed is not supported there. **(ii) The capture threshold.** From rent > acquisition + P(detect)·penalty, the central's threshold $\lambda^*_C = (k_c + p_c \cdot f)/C$ falls toward zero as its detection shrinks, while the distributed's $\lambda^*_D = k_d + p_d \cdot f/C$ is *floored* by the equal-wallet acquisition term k_d; the signed threshold gap $\delta(C) = \lambda^*_D - \lambda^*_C$ therefore stays positive across the plausible range, widening as the wallet floor k_d comes to dominate at higher project cost. **(iii) The detection floor.** With snowball detection $P = 1 - (1 - q)^m$, beating a central rate p_c needs, in the small-q (Poisson) approximation $(1 - q)^m \approx e^{-m \cdot q}$, only an expected $m \cdot q \geq -\ln(1 - p_c) \approx 0.1$ reporters — the exact Bernoulli condition is $m \geq \ln(1 - p_c)/\ln(1 - q)$, which depends on m and q separately, not only on their product. It is a model-internal detection floor under the stipulated parameters (sensitivity-dependent; see Finding 4), not an empirical burden-of-proof inversion. Three invariances bound the arbitrary-magnitudes worry —

as properties of the noise-free, large-set *expectation*, not of every finite-sample run: the advantage is invariant to the units of value (scale); in expectation it depends on the voice-bias β rather than on the participation *level* (though in finite samples turnout changes sample size, sampling variance, and hence rankings and portfolios); and, because only the first moments S^+, S^- enter the expected ranking, the Gaussian valuation draw is a convenience there rather than a load-bearing assumption (finite-sample tails and valuation shape can still move rankings). One honest boundary the runs mark: the parity law is the large-set limit; when a project's interested set is very small — a handful of people — the distributed's sampling variance dominates and a full-census central regains the edge. Two further boundaries are honest to state. The comparison is *static* — a single allocation round — whereas real harms surface over iterated cycles and feed back through elections and audit, so a persistently blind centre is the stress case, not an inevitability. And the distributed arm is *scored on the true value it reveals*, which would be circular were it not that the β voice-bias and the capture frictions make its revealed signal a biased, contestable estimate of that value rather than a definitional one.

Positioning. The first-best preference-aggregation mechanism — Vickrey–Clarke–Groves (Vickrey 1961; Green and Laffont 1979) — is infeasible for public budgets (it is not budget-balanced), so both the central planner and Core v0 are *second-best* institutions (Lipsey and Lancaster 1956); the comparison asks which second-best delivers more, not whether either is optimal. Core v0 accordingly claims not strategy-proofness — impossible for any non-dictatorial mechanism (Gibbard 1973; Satterthwaite 1975) — but *capture-resistance under bounded organized coordination*. The equal-per-citizen wallet places it in the intensity-expressing voting family (Casella 2012; Lalley and Weyl 2018) with a sharper anti-plutocratic property: it caps influence by *equal endowment* rather than by convex pricing, so money can persuade wallet-holders but cannot buy wallets — exactly the acquisition-cost floor k_d of the capture threshold. Finally, the aggregation advantage is the Condorcet (1785) jury logic and dies under its independence condition (Austen-Smith and Banks 1996): organized capture is the correlated-error violation of that independence, so the integrity layer exists precisely to defend the assumption on which distributed aggregation rests. The value primitive follows Sen's (1999) capabilities for *what* is aggregated — freedoms, not money-utility — while the *summation* rests on Samuelson (1954), an aggregation Sen himself resists; we invoke each only where it applies.

Calibration. The magnitudes are model-internal, and the gap to data defines calibration targets rather than field estimates. The central's 44–85% of benchmark-achievable value is a **candidate validation target requiring an explicit construct mapping** — not a direct check: the ex-post realized-to-appraised ratio (the World Bank's Independent Evaluation Group ratings; Flyvbjerg, Bruzelius and Rothengatter 2003) is a *different construct* from central selection relative to a full-information benchmark, so bridging them needs a stated mapping before either can calibrate the other; the voice bias β can likewise be anchored to measured participatory-budgeting demographics, and to the documented over-representation of motivated, unrepresentative participants in open processes (Einstein, Palmer and Glick 2019), rather than assumed. And independent field evidence points the direction the model does: participatory budgeting in Brazil shifted spending toward sanitation and health and lowered infant mortality at constant per-capita budget (Gonçalves 2014) — a real-world instance of citizen-directed allocation delivering more real value, cleanly separable from any magnitude the model reports. The calibration-targets appendix makes the model-internal / data boundary a visible line.

Finding 5: selection and delivery compose multiplicatively — the architecture's advantage is delivered value, not selection alone. A fifth experiment (`scripts/simulation/e4-v5/e5-delivery.mjs` , rebuilt on the clean E4 engine) adds the execution stage the first four omitted, as an **in-**

dependent delivery regime crossed with the two selection regimes — a four-cell design so each layer reads separately and jointly on the *same* funded portfolios. Executors have hidden types: an intrinsically honest share deliver; the rest divert whenever a temptation draw beats the regime's deterrent `p·[(1−a(1−r)) + γ + R]` (detection p, advance exposure a, recovery r, guarantee γ, reputational stake R) — Okun's (1975) leaky bucket. The **opaque** status-quo regime's emergent value loss is moment-matched to Olken's (2007) ~24% missing-expenditure figure (not identified as welfare); IMF's (2015) ~30% public-investment inefficiency is a broader process loss, and Reinikka and Svensson's (2004) ~87% Ugandan capture is a tail, not the central case. The **verified** regime is the architecture: a milestone-gated advance plus a performance guarantee, retention, recovery, and a reputational stake — magnitudes declared, directions from Propositions 1–4.

Every cell is a percentage of the same full-information greedy reference at perfect delivery (a heuristic normalizer, not an optimum), so no compound multiplier is reported. Selection efficiency reproduces E4 (distributed ≈ 98%, central ≈ 44% of the reference); delivery efficiency is ≈ 78% opaque versus ≈ 95% verified. Read as two main effects at the declared world, the delivery layer adds ≈ 8 points under central selection and ≈ 17 under distributed; the selection layer adds ≈ 42 points under opaque delivery and ≈ 51 under verified. The interaction is positive: the two layers **compose multiplicatively** — an accounting identity (delivered value = selected value × delivered fraction, applied per project), of which the positive interaction is the level-effect signature, not an independent discovery. The full **selection-and-delivery** architecture beats the status quo by ≈ +58.6 points of the reference (95% conditional Monte-Carlo interval [+58.0, +59.2], reflecting inner simulation variability only — world, model-form, and calibration uncertainty are not included). An earlier version summarized this as a single compound value-per-budget multiplier; that compound is **retired**, and E5 reports the layers as separate percentages.

Two refinements survive scrutiny. First, Core v0's distributed coverage is not only a selection signal: the citizens who routed the budget also observe delivery. But community coverage credibly lifts *detection*, not *recovery* (clawback needs institutional follow-up), so the coverage-only delivery dividend is small (a fraction of a point in the weak-control regime; Björkman and Svensson 2009, with failed replications; Molina et al. 2016); the sizeable delivery gain comes from the **formal** recovery channel — the verified regime — not eyeballs alone. Second, the verified regime's diversion is **low but nonzero** (≈ 2% incidence, ≈ 7% without the reputational stake): a grand-corruption temptation tail keeps a residual that strong control does not eliminate, matching Olken's finding that audits cut leakage without erasing it (2007; Avis, Ferraz, and Finan 2018; Becker 1968) — ex-ante deterrence, not an empirical zero. Within the PROBABLE world, the result is stable across seeds and under **cost/size-correlated** delivery risk (project cost independent of modeled value), and a joint delivery-parameter sweep, conditional on the declared world, keeps coverage ahead across the sampled space. Delivered value here is measured at *equal budget*; the administrative *cost* of running each institution is a separate layer (Finding 10).

Finding 6: visibility sustains the standard; naive reputation markets concentrate faster than they select. A sixth pre-registered experiment (`research/e6-reputational-competition-design.md`) isolates the incentive channel from deterrence entirely — a career-concerns setting in Holmström's (1999) sense: an all-honest executor pool with adjustable effort and no possibility of diversion (the model prices no explicit effort cost; cost-minimization is encoded behaviorally as the opaque regime's decay rule). In the opaque regime, effort collapses toward cost-minimization (0.49 → 0.24 over twenty-four cycles) — not through malice, but because effort has no return and no visible standard exists to imitate. Making verified quality visible sustains effort near its starting level, and the visible-competitive regime produces a material visibility-driven delivery gain over the opaque baseline — a pure incentive gain with zero diversion in the model. Two pre-registered predictions failed informatively. Visibility alone carries most of the effect: the mirror precedes the market (the behavioral rule ties imitation to visibility, so part of this is by construction — but the construction encodes the claim that opacity erodes professional norms by removing the standard itself). And naive reputation-weighted assignment concentrates work dramatically (assignment Gini 0.84 versus 0.34) while tracking true ability only weakly — early-luck lock-in, the cumulative-advantage dynamic of information cascades reappearing in the executor market. The design lesson runs in both directions: verified visibility is where the quality incentive lives, and any strong reputation weighting — human or algorithmic — needs the concentration observability, entrant floors, and opportunity-normalized reputation the corpus prescribes. In Core v0, reputation informs funders' choices rather than automatic assignment, with concentration visible by construction — and it never excludes: no protocol rule bars a funder from choosing any admissible actor on reputational grounds.

Finding 7: an audit-parameterized baseline — what it does and does not calibrate. The manuscript-review round's sharpest attack held that the zero-control baseline is a caricature — real administrations run audit institutions, retentions, bonds, and inspection — and the answer was a seventh pre-registered experiment (`research/e7-calibrated-baseline-design.md`) with a committed withdrawal condition: if the headline collapsed against a fair baseline, it would be withdrawn, not requalified. The audit-parameterized status-quo arm draws its parameters from published audit-institution findings (a documented-practice-informed scenario; verification of some primary sources is ongoing) — detection from Chile's comptroller works studies, retention from documented payment-state practice, recovery from Mexico's ASF series, leakage anchors category-matched to construction (Olken 2007; the multi-country evidence base spans the U.S. GAO, the U.K. NAO, the European Court of Auditors, Brazil's TCU and CGU, and the comptrollers of Chile, Peru and Colombia; Ferraz and Finan 2008) — with the planner's inspection bandwidth scaled to scope and coordinated signal bias swept as the Condorcet failure regime. The withdrawal condition was not triggered *within this apparatus*: against the audit-parameterized baseline the earlier compound was substantial at scale but only near-parity at municipal pilot scale (10-40 projects), where central selection with full coverage is competitive and the case rests on delivery and metering. But the audit evidence *parameterizes the baseline's leak*; it does **not** calibrate the Core v0 institutional treatment effect, which is governed by the later pre-registered symmetric test (Section 6), whose verdict was NO-GO — so these E7 figures are retained as conditional apparatus outputs, not a surviving headline. Within this apparatus, and conditional on its stipulated opportunist-cost distribution and no-memory baseline, one qualitative result is instructive: at the model's audit-reported detection intensity (primary-source verification ongoing), without reputational memory, the model deters no diversion — the audit-parameterized regime's incentive threshold lies below every modelled opportunist's cost, so its leak equals the zero-control regime's, and in the model the added detection reduces the visibility gap (from twenty-nine to nineteen points) rather than raising delivered value. These are model-internal outputs of E7's stipulated apparatus, not an estimated causal effect of real-world auditing. The audit-parameterized arm's leak lands inside the audit-reported leakage band (24-48% in works); the model's leak mechanics, fed audit parameters, are *consistent with* the documented range — this parameterizes the baseline's leak, it does not calibrate the institutional treatment effect. And the bias sweep bounds the open-construction claim honestly: distributed selection degrades near-linearly with coordinated signal capture and crosses below a competent full-coverage municipal planner only at roughly a thirty percent coordinated share — it degrades, never collapses, and remains superior everywhere below ten percent.

Finding 8: the architecture ranking survives endogenous behavioral participation. A pre-registered eighth experiment (E8, `research/e8-behavioral-participation-design.md`) then replaced the participation side of these arms — the default share and informed share the architecture arms had imposed — with adoption trajectories generated by a companion behavioral study: a Core v0-conformant agent-based model of awareness, registration, participation modes, and trusted microdelegation, calibrated with LLM-elicited synthetic priors (replication package: the distributed-governance-experiments repository). The architecture ranking is invariant across the three synthetic populations and all scales, including a launch trajectory that begins near zero participation — where the default layer anchors the thin early cycles by construction. The behavioral study also independently reproduces the informed-share assumption these experiments had imposed: 0.309 emergent against the 0.30 assumed.

Finding 9: the full stack — planning, selection, and delivery — and an honest accounting of what each layer contributes. A ninth experiment (`scripts/simulation/e4-v5/e9-fullstack.mjs`, built on E5) adds the third architectural layer, **planning** (constructing the eligibility frame and per-sector budget shares): E9 compares central and distributed versions of all three layers in a 2×2×2 factorial applied across ten persistent sectors, the COFOG count (United Nations 1999). A Shapley attribution decomposes the all-distributed-versus-status-quo gap into layer contributions that sum exactly to it. In the declared PROBABLE world the full stack beats the status quo by ≈ +57 points of the reference [95% conditional Monte-Carlo interval +56.8, +58.1], and the conditional Shapley split is planning ≈ +3, selection ≈ +43, and delivery ≈ +11 points. Two honest qualifications govern the reading. First, the attribution is *conditional*: every layer value is computed through the declared planning sector generator, so the standalone, quantified **selection** and **delivery** figures are the E5 ones (no planning layer); E9 contributes the three-layer *structure* and the attribution *method*. Second, **selection and delivery are large in PROBABLE but not robust across the four named worlds** (PROBABLE, PRO_CENTRAL, MYOPIA_OFF, and PRO_DIST): selection turns negative in PRO_CENTRAL, and delivery turns negative in PRO_DIST because stronger delivery magnifies a harmful portfolio. Despite those component reversals, the full Core v0 diagonal remains positive in all four — a fact the paper reports rather than hides. The named-world decomposition (conditional Shapley attributions through the declared sector generator, points of the reference):

world	full-stack gain	planning	selection	delivery
PROBABLE	+57.1%	+3.1%	+42.7%	+11.3%
PRO_CENTRAL	+14.7%	+1.5%	−2.8%	+16.0%
MYOPIA_OFF	+44.7%	+2.5%	+29.5%	+12.6%
PRO_DIST	+172.6%	+4.8%	+169.1%	−1.4%
The **planning** layer's value operates chiefly through **agenda capture**				
— the central keeping whole high-need, low-visibility functions off the				
menu (the second face of power; Bachrach and Baratz 1962; Schattschneider				
1960). That mechanism is real and its *direction* is anchored (the COFOG				
taxonomy; the pre-election shift toward visible spending, Drazen and Eslava				
2010; the systematic neglect of maintenance and prevention, Rioja 2003),				
but its *magnitude* cannot be identified without country-specific budget				
data. We therefore report planning only as the **conditional** Shapley				
contribution in the table above (≈ +3 points, small and positive across the				
named worlds), **not as a standalone anchored figure**: quantifying it as a				
standalone effect with the mechanism switched off would understate it, and				
switched on it is not yet anchorable. Chile provides a qualitative illustration only:				
mental-health spending remained near 2% of health spending over 2007–2015,				
despite mental disorders being the country's leading cause of disability				
(Errázuriz et al. 2015). That pattern is				
consistent with low-salience under-provision, but it neither identifies				

world	full-stack gain	planning	selection	delivery
agenda capture nor calibrates its magnitude; mental health is a funded				
subfunction, not an excluded top-level COFOG function.				

Finding 10: administrative cost favors Core v0 — roughly neutral only under a conservative low-spread floor, a declared advantage under an asymmetric-cost scenario. A tenth experiment (`scripts/simulation/e4-v5/e10-costs.mjs`) adds the administrative and machinery costs each institution operates and Core v0 largely replaces — the value-proxy studies, allocation and prioritization apparatus, and licensing the central carries, against Core v0's own platform and control machinery. The model handles these costs in three ways. It subtracts them from each arm's *budget* before selection, so the value loss is sub-proportional because greedy funding cuts the marginal projects first; it charges Core v0's own verification and recovery machinery rather than treating it as free; and it avoids double-counting the delivery leakage already modeled in E5. We report two **declared** scenarios over identical cost perimeters (steady-state régime costs only — one-time implementation is **CAPEX-excluded, not amortized**). Under a **conservative low-spread floor** ($\kappa\_C = 0.08$, $\kappa\_D = 0.05$, both inside a 1–10% administrative-overhead band) the layer is roughly neutral: administrative cost moves the modeled gap from +58.6 to +57.7 reference points, a −0.9-point contribution. But that low spread is a deliberate floor, not the only defensible case. In steady state the central's machinery — appraisal and value-proxy studies, the prioritization and allocation bureaucracy, approvals, and the salaries that run them — is plausibly a large recurring overhead, while Core v0 runs on a digital platform plus AI-assisted and citizen-sourced fiscalization at near-zero-marginal cost. A second, **asymmetric-cost scenario** ($\kappa\_C \gg \kappa\_D$) is therefore declared, and it splits into two statements. First, an **assumed net-budget difference**: under a declared $\kappa\_C = 0.12$ versus $\kappa\_D = 0.02$ the central's modeled steady-state operating cost exceeds Core v0's by about a tenth of the budget — a declared scenario input, not a measured saving (and $\kappa\_C = 0.12$ reads as recurring operating cost broadly — appraisal, prioritization, salaried bureaucracy — not pure overhead, since it sits just above that 1–10% band). Second, its **delivered-value effect** is smaller (about +0.4 reference points on the gap) because that freed budget funds marginal, low-value projects (the net-budget sub-proportionality). So the honest reading is not "no cost difference": the **direction favors Core v0**; "roughly neutral" holds only under the conservative low-spread floor; and the declared asymmetric-cost scenario is a declared — though welfare-modest and unquantified — cost advantage. The cost shares are declared scenario inputs, not measured arm-specific costs: the central-machinery direction reflects the large, recurring appraisal/prioritization bureaucracy documented in public-administration cost accounts; the platform side reflects the low operating cost of established public e-procurement systems (KONEPS, ChileCompra, ProZorro) and near-zero marginal AI/citizen fiscalization.

What survives. Stripped to essentials: (1) a pre-registered decision test (§6, Appendix E4) returned NO-GO — on a symmetric benchmark the distributed selection advantage was *positive but small* (median $\Delta = 0.025$, below the 0.05 rebuild rule), and its registered consequence was to retire the quantitative multiplier and treat the simulation as an illustrative frontier; (1b) a v1.14 four-scenario robustness extension (a *separate, exploratory* analysis under a different data-generating process, not a reclassification of that gate) models the central as the evidence *directionally* describes it — *near-blind to diffuse harm on the low-visibility long tail* (Hayek 1945; Scott 1998; Olson 1965; Bandiera–Prat–Valletti

2009): in that source-motivated declared reference scenario coverage-routed selection recovers *substantially* more of the model's greedy reference (≈ 98% vs ≈ 44%); even a central granted the full harm-aware competent bundle still loses (≈ +14%), and the central pulls narrowly ahead only by abandoning the declared premises — on a near-harmless world — while the *mirror* idealization of the distributed arm wins by a landslide, all a conditional model contrast reported as declared reference points, not calibrated impact; (2) the load-bearing contributions are the architecture and the qualitative credit-versus-coverage mechanism — credit-pressured central ranking underweights diffuse value that coverage-based distributed selection surfaces; (3) every compound value-per-budget ratio an earlier version reported is retired here as a calibrated effect — it was a conditional, model-internal apparatus output, not a calibrated field effect; and (4) taken together, E5–E10 locate the conditional advantage in selection and verified delivery, not overhead: E5 measures those margins separately on matched portfolios; E9 supplies a conditional full-stack attribution whose component signs vary by world while the full diagonal stays positive; and E10 leaves administrative cost roughly neutral under a conservative low-spread floor, with the direction favoring Core v0 (a declared advantage under an asymmetric-cost scenario). Planning remains unresolved quantitatively — its agenda-capture mechanism is directionally grounded, but its standalone magnitude is deliberately deferred rather than manufactured from an unanchored scenario. Any calibrated total delivered-value effect on real data remains future work.

7. Adversarial review as method

The architecture was developed under a documented adversarial loop: **attack** (a brief stating a failure mode, its location in the corpus, a stress scenario, and literature anchors) → **paired defense** (an objective evaluation classifying the attack as founded, partially founded, or a difference of judgment, with line-anchored citations into the corpus) → **resolution** (an accepted document that either integrates a mechanism or bounds the risk) → **propagation** (the resolution's constraints threaded through every affected architecture document). The loop ran five rounds, all resolved; every resolution is propagated through the corpus except the manuscript-review round's four paired defenses (D037–D040), which carry accepted resolutions whose remaining corpus propagation is tracked in the remediation roadmap. The first round: eighteen attacks on the architecture's mechanisms (metric gaming, fiscalizer capture, disbursement gaming, collusion, related-party control, complexity, incumbent resistance, among others). The second: fifteen deliberately deeper attacks on the political and behavioral foundations (democratic mandate, agenda-setting, fiscal dependence, thin markets, meta-governance vacuum, rational ignorance, cascades, clientelism, polarization, intertemporal myopia, the problem of many hands (Thompson 1980)). The third round emerged from the method turned outward: a simulated five-profile external review of this paper's companion document generated reviewer questions the corpus could not answer with existing anchors, and the standing rule converted the two serious ones into formal attacks — the legal characterization of the citizen allocation act, and the administrative capacity cost of tutored operation — both since resolved and propagated. The fourth round turned the same instrument on this manuscript itself: five simulated reviewer profiles (academic, public law, systems architecture, public-sector practice, educated general reader) attacked the published v1.6, and their five unanswerable objections became formal attacks, each now resolved — the zero-control baseline as a calibration strawman (answered by the seventh experiment and a binding reporting rule), the reserve of law over the allocation competence (an enabling-norm record gating binding mode), reputational exclusion as an unprocessed sanction (reclassified: the design holds no exclusion power to sanction with), allocation traceability against preference secrecy (resolved as citizen allocation secrecy with public money), and the adoption paradox (an adoption layer under an explicit thesis boundary). The fifth round was an ablation round of three attacks (A041–A043): the piecewise deployment of the deterrence stack, the unregulated budget-release valve, and the verification layer under machine-and-ring collusion — each resolved and propagated (`docs/111` – `docs/113`). The method's honesty requirement applies to itself: several resolutions answer their attacks with an explicit "bounded, not solved," and the full review record is public.

The loop terminates by the integrate-or-bound rule (P007). Its output discipline is what distinguishes it from ordinary threat modeling: every bounded attack must leave three artifacts — an explicit boundary sentence ("Core v0 does not require X"), a visible residual risk, and a one-sentence limitation statement. The limitations section below is therefore not a gesture of humility; it is the accumulated, adversarially generated output of the method. Of the forty-three attacks, none was dismissed; nine of the second round's fifteen were classified founded outright and the other six partially founded, all five of the manuscript round were classified founded at least in part, and the corpus's answer to several is an honest "bounded, not solved."

We used the loop with a single design team plus AI assistance — which is why we call it structured self-critique rather than validation; a self-administered adversary, however disciplined, cannot substitute for independent attack. Its obvious next application is with genuinely independent reviewers, which we identify below as the first item of future work.

8. Limitations

Stated per the method's own rule — each is a recorded boundary with a named residual risk.

Constructing the eligibility frame is centralized in the transition modes. In the closed and tutored operating modes Core v0 specifies for pilots, the implementing authority constructs planning scopes; the architecture makes that construction public, versioned, mandate-bearing, and contestable through visibility, but in those modes it does not distribute it. Constructing the scope means defining the frame — which purposes, which budget share, which protected floors, which admissibility rules — not designing or ranking projects: project creation and prioritization remain distributed even in tutored mode, so this residual agenda power is the power to decide what *may* be funded, never what *is* funded. It is important not to misread our own simulation here. What that simulation shows dominating every other quality margin is the informational quality of the **project prioritization** — the aggregated allocation profiles the funded share follows — and that prioritization is distributed by design, even in tutored mode; the result is therefore an argument *for* distributing construction, not evidence that a central agenda governs delivery. The residual centralized power is the narrower one: constructing the eligibility frame is itself the second face of power (Bachrach and Baratz 1962; Schattschneider 1960; Lukes 1974) — the power to keep something off the menu — which the architecture answers, in these modes, by making the frame public, versioned, mandate-bearing, and contestable rather than by distributing it. Three things scope the limitation honestly. It is a property of the transition modes, not of the architecture: operating modes are country-configured states, and the designed trajectory is open, socially constructed agenda-setting. It is narrower than the passive share: engaged citizens allocate manually, delegate, or adopt configurable profiles, so authority weights fully govern only the never-engaged share. And it is now measured rather than assumed: E4 shows open construction of the weights from aggregated citizen signals is viable and scale-robust in the model, so the constraint is no longer whether distributed construction can work in principle but whether an elicitation mechanism can keep dispersed signals honest, unbiased, and representative under gaming, clientelism, and expressive-allocation pressure — a design problem the corpus gates rather than assumes away. The comparative baseline also belongs in this paragraph: under the current institutional model, the entire budget follows a centrally constructed vector that is neither published, versioned, pluggable, nor overridable by any citizen. The transition modes reproduce that centralization visibly and revocably; the open mode is designed to end it. This remains the architecture's principal open problem, now with a measured prize attached to solving it — and for that reason we treat it not as one limitation among many but as the research program's next object: the design of open agenda-setting, including the candidate architecture in which a distributed agenda is constructed in parallel to the authority's own and the tutored role narrows to admissibility review of conflicts between the two, is the natural subject of a dedicated follow-up study and pilot.

Procedural legitimacy is not democratic mandate — and the enabling norm does not yet exist. The platform records the external authorization for budget migration and allocation formulas (the Allocation Mandate); it cannot manufacture authorization the law never granted. In the continental tradition of the reference jurisdictions, binding citizen allocation requires an enabling instrument of sufficient rank that no current statute supplies — the regional precedents (Peru's participatory-budgeting statute, Brazil's city-statute framework) prove the instrument is achievable, not that it exists — so the architecture's lawful deployments today are consultative and tutored, in which every material allocation decision remains imputed to the competent authority as a reasoned public resolution; the delivery, metering, and reputational-memory results operate unchanged under that status, and only the mature open mode requires binding allocation. The normative debate over substituting atomized allocation for representative appropriation (Rosanvallon 2008; Urbinati 2014) remains open. Under deep evaluative disagreement, the architecture's posture is procedural in Gaus's (2011) sense: its rules aim to be justifiable from diverse moral standpoints — which is what mandate records, motive neutrality, and the comparative-critique discipline provide — without presupposing a shared comprehensive doctrine. One further objection is deliberately out of scope: the model takes the coercively raised budget as given and improves its administration; the libertarian challenge to the taking itself (Nozick 1974) is neither answered nor begged here. Contribution-weighted allocation formulas, in particular, are flagged by the architecture as plutocratic departures requiring explicit higher authorization.

Fiscal dependence is measurable, not enforceable. The state controls the budget spigot. The Fiscal Commitment Profile converts strangulation from invisible to attributable — delivery latency, unexecuted valid orders, mid-cycle share cuts all become public data — but no software compels a sovereign to pay (Kydland and Prescott 1977; North and Weingast 1989). Credible commitment must come from country-level law.

Verification quality is assumed, then priced. Propositions 1–4 take detection and discovery probabilities as parameters. In thin control markets — credence-good markets where quality is unobservable to the buyer (Akerlof 1970; Dulleck and Kerschbamer 2006) — both collapse simultaneously, and the only compensating margins are financial terms and imported (remote or cross-territory) verification. The architecture prices weak verification; it cannot conjure verifiers.

Coverage assumes source diversity it must then guarantee. The distributed arm's advantage rests on many partly-independent signals; when profile providers, delegates, and recommenders concentrate onto a shared platform or a few super-delegates (Kling et al. 2015), their errors correlate and the common-mode channel — the single sensitivity that materially shrinks the anchored gap (§6, Appendix E4) — recaptures the very epistemic bottleneck coverage is meant to replace. Independence is therefore not a modeling convenience but an architectural obligation: Core v0 must publish delegate, profile-provider, and recommender concentration as observable quantities and trigger diversification thresholds when they climb. The residual risk is that a market converges on one recommender faster than the thresholds bite.

Behavioral realism cuts both ways. The simulation vindicates designing for inattentive citizens, but it equally shows that a defaults-weak deployment degenerates into salience-driven allocation. Off-platform phenomena — clientelist brokerage, expressive polarization, collusion conducted entirely outside the system — are made harder and more discoverable, never impossible; the architecture's claims are comparative (against opaque monopoly), not absolute.

Meta-governance in open mode is deferred by design. Rule-change procedure, versioning, and non-surprise constraints are specified; the constitutional mechanics — rules for making rules (Buchanan and Tullock 1962) — of who votes on protocol changes in a mature open-mode deployment are deliberately not. Open-mode deployment is gated on resolving them.

Adoption selects, and the thesis does not depend on it. This paper addresses whether the architecture can be built and reports conditional model contrasts for both selection and delivery — not whether Core v0 delivers more value in the world (a target-domain total delivered-value effect remains a separately identified future estimand, not claimed here), and not whether any authority wants it. The corpus supplies the deployment layer for an authority that has decided (prospective baselines measured from instrumentation onset, credit attribution on verified delivery, institutional rather than personal timeout attribution in the first cycle, and a symmetry clause barring any operator from exempting its own projects), and names the plausible adopter archetypes — the post-scandal challenger, the mandating higher government, the conditioned external funder. The honest selection effect stands: the architecture will plausibly be adopted first by relatively clean or newly arrived sponsors, in the places that need it least.

The layered magnitudes are conditional, not portable. E5's ≈ +58.6-point selection-and-delivery estimate [95% conditional Monte-Carlo interval +58.0, +59.2] holds the declared world, model form, and calibration fixed; the interval reflects Monte-Carlo variation over simulated worlds, not uncertainty about the data-generating process, calibration, functional form, literature transport, or field implementation, and the greedy reference is a heuristic normalizer, not an optimum. E9's Shapley values are conditional attributions through the declared sector generator, not standalone estimates; individual layer signs reverse in extreme worlds even though the full Core v0 diagonal remains positive in all four named worlds. Planning's agenda-capture direction is anchored but its magnitude is unquantified. E10's cost result rests on declared net-budget shares (scenario inputs, not measured arm-specific costs): roughly neutral under a conservative low-spread floor, and a modest Core v0 advantage under a declared asymmetric-cost scenario ($\kappa\_C \gg \kappa\_D$). None of these is a field-effect estimate.

The outcome measure is a bounded, non-distributional aggregate. The value the model scores is a cardinal utilitarian-style sum over affected parties on a bounded slice of project-shaped, milestone-verifiable public investment (infrastructure is the clearest case, but the architecture is not limited to it); it is not distributionally weighted and says nothing about redistribution, equity, or who bears benefits and harms. A portfolio can score well on this measure while distributing badly. Applying the criterion across groups, or to the whole budget or the purpose of taxation, would require a separate social-welfare and incidence specification this paper does not provide. The model is also partial-equilibrium: strategic reporting, endogenous proposal supply, project complementarities, tax incidence, and general-equilibrium effects are outside it (see the claim-and-estimand-contract).

Epistemically, this is one team's self-critiqued design. The adversarial corpus was produced by the same research effort it attacks, with AI assistance; the earlier apparatus's status-quo baseline was audit-parameterized (its parameters drawn item-by-item from published audit-institution findings, with primary-source verification still in progress), while the current symmetry gate is an uncalibrated stylized selection test — neither is calibrated to a specific PB dataset, and the remaining parameters are plausible rather than measured; and no pilot has been run. The three missing validations — independent expert attack, calibration to empirical PB data, and a bounded tutored pilot (sports-sector, one municipality) — are the research program's next phase, in that order.

9. Implementation pathway

The architecture is built for gradual, revocable adoption, and this section is explicit about what it does and does not claim: the paper's question — can the two-hundred-year-old allocation architecture be re-architected with today's technology — is answered at the level of a buildable design and a conditional selection-mechanism *direction*, independently of whether any authority chooses to deploy. It does **not** answer *by how much* real-world delivered value would improve: that is a separately identified estimand left to independent review, empirical calibration, and a bounded pilot (§8). What follows is the pathway for an authority that chooses to deploy. A country opens one public function (the reference pilot is municipal sports infrastructure), migrates a small budget share under a tutored operating mode, and retains admissibility review — with every tutored decision and delay public by construction. The pilot's default framing is prospective: instrumentation begins at adoption, the visibility gap is published as the adopter's declared starting line ("measure me from here"), and pre-adoption figures are reported separately, impersonally, and as context — the configuration under which exposing instruments have historically been adopted. Functional maturity metrics (participation mix, default-flow share, fiscalization independence rates, incumbent- resistance indicators, fiscal reliability) determine whether the deployment earns wider scope, and their trajectories, not rhetoric, answer whether distribution outperforms the local baseline. The exit condition is honest in both directions: a pilot whose indicators stagnate under incumbent throttling documents that fact publicly, which is itself information the current system never yields.

The transition between regimes is itself measured. The companion experiments (Offermann 2026b) quantify the semi-open regime of the operating-regime ladder (docs/110) — a bounded mandated envelope running on protocol autopilot beside the authority's traditional budget — as a fiscal blend: above a portfolio-granularity floor of roughly ten percent, blended verified value rises monotonically and near-linearly with the envelope share within that apparatus, from break-even near eight to ten percent upward — an earlier-apparatus contrast now subject to the retired-multiplier caveat (Section 6), not a calibrated endpoint. The transition from the status quo toward the open regime is a dial, not a leap: adoption can proceed in increments.

The same experiments measured a variable this corpus had left unregulated: *when* the authority releases budget into the allocation machinery. The resulting deployment rule: meter release against a work-in-progress ceiling calibrated to the delivery-and-verification pipeline's throughput and cycle time — never against the calendar. Calendar release freezes months of budget in escrow and saturates verification; and when verification capacity is scarce, no release policy compensates — verification capacity is the pipeline's ceiling before it is the anti-fraud instrument. The rule is conditional on a carryforward instrument (the semi-open envelope is precisely such a vehicle); under strict budget annuality it degenerates to within-year metering.

Finally, the technological premise that lowers participation costs on the citizen side (the AI tutor) applies symmetrically to the control side. Machine verification of protocolizable evidence classes multiplies fiscalization capacity, with humans as the permanent second instance — sampled re-verification with a published floor, seeded known-answer controls as the calibration and drift-detection instrument, auditing the verifier rather than competing with it — so that the machine's error rate remains measured and the human control profession remains funded from the control budget it relieves. Measured on a five-family panel of real models (Offermann 2026b), frontier models converge on good specificity and fraud detection on document-legible evidence while small local models are weaker, and evidence contracts that include objective comparison references (market benchmarks, duration bands, thresholds) let a strict verifier judge rather than guess. The machine layer reaches only document-legible, delivery-phase fraud — physical quality-below-spec and pre-contract theft remain fully human, so provenance attestation is tamper-evidence at capture, not court-grade proof, and evidentiary admissibility still needs custody, contradiction, and expert testimony. Contraposed citizen evidence — independent producers with interests opposed to the executor, whose anticipated existence deters diversion — keeps the watching distributed even as routine document-verification labor shrinks; but its strength equals the *independence* of the contributor layer, and a colluding ring that captures or silences it erases the effect. Cross-layer collusion is in fact the one adversary that bypasses the per-milestone deterrence and moves leakage by an order of magnitude (while the delivered-value advantage survives), so collusion resistance — verified beneficial ownership, contributor Sybil-resistance, and decentralization of the assigner and the audit-budget floor — is a first-class requirement (docs/113), not a residual caution.

10. Conclusion

For the bounded public-investment allocation problem studied here, a relevant criterion is not how faithfully an institution executes a plan but how much delivered, verified value it produces per unit of public resource (Musgrave 1959; Okun 1975) — one criterion alongside the distributional and rights constraints this model does not represent. This paper's contribution is an architecture that makes that criterion operational — and a disciplined account of exactly how far its evidence reaches. The architecture's spine is two separable questions anyone can ask. First: take the same projects, designed identically, and change only who executes and how they are watched — does the visibly audited regime with reputational consequences deliver more than the one without them? Second: hold the control layer fixed and change only which projects get funded, centrally planned or socially prioritized? In E5's declared PROBABLE world, the answer to both is yes — a verified-delivery gain and a selection gain that compose multiplicatively (an accounting identity, not an independent finding). Together, the full selection-and-delivery architecture reaches ≈ +58.6 points of the greedy reference in the declared world [+58.0, +59.2]. E9 and E10 then sharpen where the modeled advantage comes from: the full-stack diagonal stays positive across the named worlds even when individual Shapley contributions reverse in extreme corners; agenda capture is represented as planning's principal mechanism, its direction anchored but its standalone magnitude left unquantified; and net-budget accounting leaves administrative cost roughly neutral under a conservative low-spread floor, and yields a modest Core v0 advantage under a declared asymmetric-cost scenario. In this model, the architecture's advantage is delivered value, and administrative cost adds to it rather than detracting from it. Those magnitudes are conditional model-internal contrasts, not calibrated field effects. The paper therefore stands on the fully specified architecture, the credit-versus-coverage mechanism, and channel-separated findings on selection, verified delivery, and administrative cost — not on an institution-wide multiplier. The pre-registered selection gate returned NO-GO because its positive median $\Delta = 0.025$ did not clear the 0.05 rebuild bar; as registered, we retire the earlier multiplier (§6; Appendix E4). Its competent, harm-aware central and delivery parity determine the result's scope: this is a calibration decision under that comparator, not a finding that Core v0 failed. The separate exploratory map asks what follows under the source-motivated institutional asymmetries the gate sets aside; in its declared reference scenario coverage recovers about 98% of the model's greedy reference against the central's ~44%, while the central's narrow 2.3-point lead appears only at its near-harmless favourable endpoint. These are conditional scenario results, not a recalibration or a field-effect estimate. One model-internal result is worth carrying because it is about the delivery layer, not the multiplier: in the model, at E7's audit-informed (but not yet fully source-verified) detection intensity, detection without persistent consequences deters no diversion; what moves delivered value is the instrument the modelled status quo lacks — consequences that persist. Whether that holds in a real institution is a hypothesis for a pilot, not a result here; but the model is unambiguous that accountability without memory is bookkeeping.

The deeper point is Friedman's: a central administration spends other people's money on other people, the spending category with the least care for both cost and value (Friedman and Friedman 1980). This architecture does not answer that problem with exhortation; it re-plumbs the bucket. Planning remains — the guiding thread that sets scopes, floors, and mandates — but within those bounds value comes from joining dispersed selection to consequence-bearing delivery: measurable promises, conditional release, independent verification, consequences that compound into reputation, and a meter on every leak. The question this paper answers is therefore not whether states should be bigger or smaller, but whether the layers of state activity that fail through information and incentive monopoly can be re-architected to fail less — and to show their failures when they do. For one such layer we have specified a complete architecture, proved the incentive conditions its mechanisms depend on, measured selection, aggregation, and delivery in simulation against a baseline the incumbent system's own auditors supplied, and subjected the whole to five rounds of documented adversarial review with an explicit integrate-or-bound discipline. The result is deliberately modest in its claims and unusually explicit about their edges: allocation quality rides on the informational quality of whatever constructs the weight vector, whose open construction is measured viable but whose honest elicitation remains the open problem; delivered value rides on verification whose market conditions must be priced; legitimacy rides on mandates the platform can record but not create. What distinguishes the proposal is that these edges are specified, monitored, and attached to named objects — which is, we argue, what it looks like when institutional design is treated as an engineering discipline rather than an ideological one.

The comparison is conditional, not ontological. A center with credible local-information channels, organized representation of diffuse losers, and low credit pressure can approach parity. Core v0's claim is that those capacities are institutional accomplishments to be *demonstrated*, not virtues to be presumed — and that a state less dependent on what its center can see, and less able to certify its own mistakes, is worth the attempt.

Appendix E4: the symmetric gate and the four-scenario robustness map

This appendix gives the full design of the pre-registered symmetric gate (summarized as "Quantitative status" in §6) and the complete v1.14 four-scenario robustness map (headlined in §6): the scenario table, the harm-myopia decomposition, the frontier, the benchmark theorem, and the four limits.

The symmetric credit-versus-coverage gate (full methods)

Because this is the paper's one confirmatory computation, its design is stated in full here rather than only by reference. Each world holds $K = 500$ candidate projects; for each, $N = 5000$ potential participants are considered, each interested with a project-specific probability, so the interested reach is at most N and endogenous. Both arms then see the same candidate pool, the same exact costs, the same truth $net[j] = S[j] - h \cdot cost[j]$, delivery held at **parity**, and the same report noise report = v + Normal(0, τ); each funds a **greedy** set under a budget of one-third of total project cost, is eligible to fund a project only where *its own* noisy estimate of net is positive (no oracle gate), and its delivered value is scored on the projects' *true* net. The arms are symmetric except for the coverage mechanism and its matched counterparts. *Distributed (endogenous coverage):* each interested citizen reports independently with probability p if her value $v \geq 0$ and $p \cdot (1 - \beta)$ if $v < 0$ (adverse voice bias), giving $\hat{h}S\_D = \Sigma reports / p$, ranked by estimated net per cost. *Central (competent value-reader):* an appraisal budget matched to the distributed arm's *expected* total reports in that world, spread **evenly** across projects as a rounded fixed per-project bandwidth m_C = round(expected reports / K) (so the two arms' appraisal totals are equal in expectation up to that rounding); per project it samples m_C interested citizens, observes v + Normal(0, τ), and forms its own noisy $\hat{h}Net\_C = reach \cdot mean(observed) - h \cdot cost$. It ranks by score = $(1 - \lambda) \cdot z(\hat{h}Net\_C/cost) + \lambda \cdot z(P/cost)$ — its **own noisy estimate**, never the true net — where P is claimable political credit (the electoral credit-claiming and traceability logic by which visible, attributable benefits are favoured over diffuse ones; Mayhew 1974; Arnold 1990) and λ is bounded credit pressure (a *posited* pressure whose real-world magnitude must be measured, not assumed). Credit moves *ranking*, never eligibility (no knowingly value-destroying planner). The legitimate asymmetries are therefore only these: distributed reports self-route to projects citizens care about while negative stakeholders participate less, and central appraisal is spread evenly while its ranking carries credit pressure — everything else is shared. The estimand is **$\Delta = (D - C)/O$** per world, where D, C, O are delivered true net for the distributed, central, and full-information greedy arms and O is a reference level, not an optimum. The frozen grid sweeps $\lambda \in \{0, 0.1, 0.2, 0.3\}$ ($\lambda = 0$ a negative control), a latent-correlation setting $\rho \in \{0, 0.5, 1\}$ (realized corr(S, P) ≈ 0.00, 0.30, 0.82), and $h \in \{1.5, 2.5, 4\}$ over 100 seeded worlds, in a baseline observation regime ($p = .35$, $\beta = .30$, $\tau = .5$) and a matched-budget low-information stress regime ($p = .15$, $\beta = .60$, $\tau = 1.0$). The **pre-registered decision rule** — frozen before running and designed by the independent auditor to be adversarial — required, for a GO on rebuilding the quantitative engine, at least 15 of the 18 primary cells with mean $\Delta > 0$, a pooled **median $\Delta \geq 0.05$**, a bootstrap lower bound > 0, and median $\Delta \geq 0$ under the stress regime, plus a guard to pause if the $\lambda = 0$ control itself exceeded 0.05. The result was **NO-GO**: the advantage was positive in all 18 primary cells, but the pre-registered pooled **median $\Delta = 0.025$**, below the 0.05 rebuild gate; the $\lambda = 0$ negative control sat at ≈ 0.016, within the pause guard (no hidden asymmetry flagged). A **post-hoc** world-cluster ratio-of-sums estimate was $\Delta = 0.026$ [0.023, 0.029] (Monte-Carlo uncertainty on the simulated data-generating process, reported separately from the median). The advantage rises with credit pressure λ and falls as credit aligns with value — the credit-versus-coverage mechanism — but it is small, which is why the calibrated multiplier is retired and the simulation treated as an illustrative frontier, and the paper rests on the architecture and the mechanism direction, now sharpened by the robustness map below.

The four-scenario robustness map (v1.14)

The pre-registered gate above equips the central arm with competent, *harm-aware* appraisal. A v1.14 extension asks the empirically-grounded question: what happens when the central is modelled as the evidence describes it — **near-blind to diffuse harm on the low-visibility long tail**? That near-blindness is over-determined by the literature: the knowledge problem (Hayek 1945), state legibility (Scott 1998), diffuse costs politically under-weighted (Olson 1965; Schattschneider 1960; Wilson 1973), 83% of government waste being *passive* rather than chosen (Bandiera, Prat and Valletti 2009), the seen-versus-unseen (Bastiat 1850), and agenda control (Bachrach and Baratz 1962). The model realizes it as a salience-gated harm term, $M\_C = a + b\cdot S^{+} + w\cdot(v\_p - S^{+}) - b\_H\cdot s(V)\cdot H + \eta$, whose harm-detection $s(V)$ rises with a project's visibility, while the distributed arm registers harm across the whole distribution. **Net-allocation participation is universal by architecture (p = 1)** — in Core v0 profiles and delegates allocate on behalf of the passive, so it is a *fact*, not a low participatory-budgeting turnout. Its *signal quality* is an anchored composition: ~5% active direct reports (the single-digit turnout figure), ~35% microdelegation (individual signal, revocable — Kling et al. 2015), and ~60% profile rules (a high-alignment category default, since people overwhelmingly keep defaults — Samuelson and Zeckhauser 1988). Full literature anchoring of every calibration knob is in `research/e4-calibration-literature-anchors.md`. Scoring delivery on true value, four declared scenarios (plus one diagnostic contrast) map where each institution stands, as the signed fraction of the full-information greedy benchmark, parity at zero (`npm run e4:scenarios`):

scenario (assumptions)	m ± 95% CI	Core v0	central	winner
Probable — the declared reference scenario (central myopic to diffuse harm, projecting, credit-tilted; distributed on its anchored coverage composition)	**+54.0%** [+53.2, +54.8]	98.2%	44.2%	**Core v0 (decisive)**
Only harm-myopia switched off (`MYOPIA_OFF`; diagnostic contrast: probable, changing ONLY the two harm-gate coordinates)	**+37.6%** [+37.0, +38.2]	98.2%	60.6%	Core v0
No-myopia bundle — probable, but the central is granted harm-sight + unbiasedness + precision + no credit	**+13.8%** [+13.5, +14.1]	98.6%	84.8%	Core v0
Distributed's favourable case	**+205.2%** [+202.9, +208.1]	95.6%	−109.6%	Core v0
Central's declared favourable endpoint (a residually-imperfect reader on a near-harmless world)	**−2.3%** [−2.5, −2.2]	95.3%	97.6%	central (narrow model-internal lead)

The anchored result is large and robust across the declared space (exploratory). Under the **declared reference scenario** the distributed arm delivers ≈ 98.2% of the benchmark and the central

≈ 44.2% — a +54-point gap — and coverage wins across essentially the whole anchored parameter space. Turning off harm-myopia *alone* (the two harm-gate coordinates) recovers about **41%** of the gap (16.4 of a 40.2-point decline); granting the central the *full* competent-but-harm-aware bundle recovers the rest yet still leaves coverage ahead (**≈ +13.8%**) — so even a central idealized in every way, **harm-sight included**, still loses. The central pulls narrowly ahead only by **abandoning the declared premises**: a corrected reader (no myopia — against Hayek/Scott/Olson/Bandiera; no projection — against Broockman and Skovron 2018) on a near-harmless world reaches only ≈ −3%, a marginal, anti-empirical corner. That corner is reported **symmetrically** with the distributed arm's *equally-idealized* corner — built to mirror the same recipe: perfect distributed signal on a harm-rich world with the central kept at its *anchored* myopia — which reaches **≈ +118%** (the broader `PRO_DIST` scenario in the table, +205%, is more favourable still because it *also* degrades the central below its anchored level). Idealization is wildly asymmetric, and neither corner is empirically grounded. The one sensitivity that materially shrinks the anchored gap is **correlated / common-mode error** on the profile-and-delegation share (a shared platform/recommender, or delegation concentrated on super-delegates — Kling et al. 2015): it takes ≈ +54% down to ≈ +44% (modest) and ≈ +26% (strong), crossing parity only at a large shared-error level (σ_cm ≈ 2.1). No single-factor slice flips the winner over its plausible range; the combined ceteris-paribus path from the declared reference to the fully-idealized central endpoint crosses parity only at **t ≈ 0.92 of the declared segment**. This is local, ceteris-paribus curve evidence — six one-factor slices plus one declared combined path — not an exhaustive global frontier; a joint Sobol / Latin-hypercube sweep of the full parameter space is deferred to future work. These magnitudes are **declared, source-motivated reference points from a stylized comparative- institutions model — a conditional model contrast, not target-domain calibrated field effects**. The standing limits are: (i) the harm-gate's exact *magnitude* is a stylized functional form — its *direction* is strongly anchored, and the result is robust across the s_exp ∈ [1, 2.5] band (≈ +48% to ≈ +54%); (ii) the central inputs carry an unpropagated transport gap — the political-opinion evidence identifies elite–constituent *perception* error, and mapping it to project-level welfare error requires three unestimated links (opinion misperception → project scoring → portfolio choice → realized affected-party value), so those inputs are proxy-informed, not calibrated; (iii) the reported intervals are 95% conditional world-bootstrap intervals at *fixed* scenario inputs — finite-world simulation uncertainty only, excluding uncertainty in parameter values, literature transport, functional form, and field implementation; the distributed arm's independent-plus-common-mode error is its one structural sensitivity (above); and (iv) delivery and administrative cost are handled in separate experiments — E5 adds delivery (leakage) at *matched budget*, and E10 adds administrative cost as a *net-budget* reduction before selection — not folded into this selection map. The reproducible scenarios, frontier, evidence, theorem, and full literature anchoring are in `scripts/simulation/e4-v5/`, `research/e4-parity-theorem.md`, and `research/e4-calibration-literature-anchors.md`.

E4 calibration targets

The E4-v4/v5 magnitudes are model-internal; the table names, for each parameter, the real dataset that *could* inform it — making the boundary between model-internal and empirically-anchored a visible line rather than a caveat buried in prose (details in `research/e4-calibration-targets.md`). The central %-benchmark is an *output* the model computes, but mapping it to observed realized-to-appraised ratios is **not a direct overlay**: the two are different constructs (§6), so it is a **candidate validation target requiring an explicit construct bridge**, not a one-step calibration.

Model quantity	Model value	Real-world proxy	Candidate dataset(s)	Status
central %-benchmark	44–85%	realized ÷ appraised value	World Bank IEG ratings; Flyvbjerg megaproject DB	candidate target; needs an explicit construct mapping
η (harm-blindness)	0–0.5	passive vs active waste share	Bandiera-Prat-Valletti 2009 (83% passive, setting-specific: Italian standardized-goods procurement)	anchored-direction
β (voice inequality)	0.2–0.5	PB participation bias	NYC / Paris / Porto Alegre; Decidim / Consul	calibratable
q, m (detection)	q ≈ 0.5–1%, m in hundreds	complaint / whistleblowing rates	FTC Consumer Sentinel; NYC 311; Dyck et al. 2010	calibratable
λ threshold	central ≈ 0.10	procurement rents / bribe depth	Olken 2007; WB Enterprise Surveys	calibratable
penalty f	equal both sides	legal sanction scale	held equal (conservative)	scope choice

The v1.14 four-scenario map (above; headlined in §6) makes the same anchoring explicit for its harm-myopia model: the visibility long tail is source-motivated by heavy-tailed public procurement (Skuhrovec et al. 2013), participation by participatory-budgeting turnout, and the harm-detection gate by the agenda-setting/salience literature; the per-knob anchors and their strength are recorded in `research/e4-plausible-anchors.md`, with the reproducible scenarios, frontier, and theorem in `scripts/simulation/e4-v5/` and `research/e4-parity-theorem.md`.

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Companion materials: formal proofs (formal-models), simulation code and full result tables (`scripts/simulation/allocation-sim.mjs`, simulation-results), the audit-institution evidence base (audit-evidence-base), the architecture corpus (`docs/`, `knowledge/`), and the complete adversarial record (`attacks/`, `defenses/`, accepted resolutions `docs/67` – `docs/113`; all forty-three attacks re-

solved and propagated, except the manuscript-review round's D037–D040, whose corpus propagation is tracked).